

RITESH PATIL

Pune, Maharashtra, India

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🌐Ritesh-Patil

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PROFESSIONAL EXPERIENCE:

Web Developer (Intern)

Jnana Prabodhini

Jan 2025 - June 2025

- Contributed to developing both offline and online versions of the Virtual Kiosk using AngularJS, enabling content delivery through smart JSON routing with .exe deployment and YouTube fallback
- Engineered a responsive UI using core AngularJS principles, enhancing kiosk interactivity and user engagement across platforms

EDUCATION:

P.G Diploma in (AI & ML)

MITWPU, Pune, India

July 2024 – May 2025

- Relevant Coursework: Machine Learning, AI, Data Mining, Neural Networks, NLP, Computer Vision, Statistical Analysis, Deep Learning: applied concepts through hands-on projects

BSc Computer Science

Abasaheb Garware college, Pune, India

Aug 2020 – June 2023

- Computer Networks, Database Systems, Algorithms, Programming Languages, Computer Architecture, and AWS Cloud

HSC

Abasaheb Garware college, Pune, India

Aug 2018 – May 2020

CERTIFICATION:

[Oracle Cloud Infrastructure 2025 Certified Generative AI Professional](#) | LLM, RAG

[GenAI & LLM Workshop](#) | LMStudio, AnythingLLM, RAG, HuggingFace, Prompting Strategies

[AWS Academy cloud foundation](#) | Skilled in EC2, S3, Lambda, API Gateway, RDS, VPC, IAM

LANGUAGES AND TECHNOLOGIES:

Languages : Python, C, R, SQL

Libraries : Pandas, Numpy, Scikit-learn, openCV, Matplotlib, Seaborn, LangChain, LangGraph

AI/ML skill: Generative AI, Agentic AI, Deep Learning, LLM, RAG, image processing

Data Analysis: Power BI, Tableau, Excel, Data Cleaning, Dashboard Design

AWS : AWS (EC2, S3, Lambda, RDS, IAM, VPC), Docker, CI/CD

PROJECTS:

Medical Chatbot (RAG, LangChain, GPT-4, Pinecone, AWS, Docker)

- Built end-to-end RAG-based chatbot using 637-page medical textbook with Sentence Transformer embeddings
- Orchestrated LangChain, GPT-4, and Pinecone for retrieval-augmented, context-aware medical query resolution
- Implemented CI/CD pipeline with GitHub Actions, containerized with Docker, and deployed on AWS EC2

Heart Disease Prediction Using Retina Images (YOLO, TensorFlow, OpenCV)

- Implemented YOLO for precise detection and classification of retina features linked to heart disease, improving detection
- Utilized Roboflow for annotating 800+ images and training data, reducing preprocessing time by 50%
- Designed and trained CNNs in TensorFlow and OpenCV, increasing marker identification accuracy by 25%

Employment Assistance Tool using LLMs, RAG, and Neo4j

- Developed an AI-powered employment tool using LLMs, improving job-role matching accuracy by 30%
- Applied **Retrieval Augmented Generation (RAG)** to speed up search and enhance recommendation quality
- Integrated APIs and Neo4j knowledge graph to generate personalized results, boosting engagement by 25%

Movie Recommender System

- Built a content-based movie app using Streamlit and TMDb API, fetching 1,000+ dynamic posters
- Recommended top 5 movies based on preferences, achieving a 95% user satisfaction rate
- Serialized ML models using Pickle to enhance speed and improve recommendation accuracy by 15%

E-commerce Product Insights Dashboard using SQL and Power BI

- Analyzed 5,000+ SKUs using SQL to identify sales trends and underperforming products
- Created a Power BI dashboard that increased decision-making speed and boosted campaign ROI by 20%
- Reduced manual reporting time by 30% with auto-refreshing SQL queries and optimized pipelines

Hospital Patient Record System

- Designed a structured SQL database to manage patient records, appointments, and billing information
- Created complex queries for patient history, doctor schedules, and admission tracking
- Improved record retrieval time by 35% through normalization and query optimization

AWS Cloud Migration - Lift and Shift Approach

- Migrated multi-tier applications to AWS cloud with minimal changes, increasing uptime to 99.5%
- Reduced cloud infrastructure costs by 15% using Auto Scaling, ELB, and resource optimization
- Managed deployment using EC2, S3, Route 53, and VPC for scalable, high-performance architecture